

Chapter 1

Angles

As opposed to being a measure, we define an angle as a geometric figure, ie: a set of points. Specifically, an angle is the union of two rays that share a common vertex.

Definition 1 (Angle). An *angle* is the union of two rays $\overrightarrow{AO}$ and $\overrightarrow{OB}$ that share a common vertex O , denoted as follows

$$\angle AOB := \overrightarrow{AO} \cup \overrightarrow{OB}.$$

Using the half-planes defined previously, we have the tools necessary to describe the interior and exterior of an angle.

Definition 2 (Interior of an Angle). For any three non-collinear points $A, B, C \in \mathbf{P}$, the *interior* of $\angle ABC$ is the set:

$$\text{Int}(\angle ABC) := \{P \in \mathbf{P} \mid \mathcal{S}(P, A, \overrightarrow{BC}) \wedge \mathcal{S}(P, C, \overrightarrow{BA})\}.$$

Equivalently, this is the intersection of two half-planes:

$$\text{Int}(\angle ABC) := H_{\overrightarrow{BC}, A} \cap H_{\overrightarrow{BA}, C}.$$

Definition 3 (Interior of an Angle). For any three non-collinear points $A, B, C \in \mathbf{P}$, the *interior* of $\angle ABC$ is the set:

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